

ROOP SAGAR MANGINENI

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Education

University of Illinois

Master of Science in Computer Science

Aug 2023 – May 2025

Springfield, Illinois

Technical Skills

Languages: Python, SQL, C++, HTML, CSS, JavaScript

Frameworks Tools: Flask, AWS, Snowflake, Retool, Spark (conceptual)

Data ML Stack: Machine Learning, Data Modeling, ETL Pipelines, NLP, Deep Learning, Large Language Models

Libraries: Pandas, NumPy, Scikit-learn, TensorFlow, PyTorch, OpenCV, LangChain

Experience

Prosrvc LLC

Data Scientist / Software Engineer

June 2025 – Present

Bothell, WA

- Built Python-based ML models to predict phone resale value and detect device defects with **87%+** accuracy.
- Developed NLP tools to extract insights from reports and reviews, reducing manual review time by **40%**.
- Developed data pipelines that process and analyze over **10K** device logs monthly, improving analysis speed.
- Designed and deployed Flask APIs integrating AI predictions into dashboards for operations and sales teams.

University of Illinois

Graduate Research Assistant

Aug 2024 – April 2025

Springfield, Illinois

- Applied text preprocessing (tokenization, lemmatization, stopword removal) to clean and prepare large textual datasets.
- Improved BERT model accuracy from **13% to 86.4%**, evaluated with precision, recall and F1-score.
- Developed matching algorithms between large datasets, increasing text-matching accuracy and reducing manual error.
- Performed exploratory data analysis (EDA) and feature extraction to optimize model performance.

Accelerize 360

Data Science Intern

May 2024 – Aug 2024

Dallas, Texas

- Integrated Salesforce data with Snowflake, mapping data types and creating structured tables from **10+ objects**.
- Built internal dashboards using Retool to automate data integration and analysis workflows.
- Developed ETL pipelines in Python and SQL to transform and schedule processing of **1M+ rows** of Salesforce data.
- Ensured high data quality and enabled business decisions through robust modeling and pipeline optimization.

Projects

Job Search Web App | [Python](#), [Streamlit](#), [RapidAPI](#), [NLP](#)

- Implemented NLP-based skill extraction with **90%+ accuracy** for relevant job recommendations.
- Integrated RapidAPI to retrieve and filter **60+** postings per query across roles and locations.
- Designed a responsive UI that cut search time by **70%** using automated filtering and visualization.

Automated Blog Generation with LLama 2 | [Python](#), [GenAI](#), [LangChain](#)

- Engineered an automated blog generation pipeline using LLama 2, reducing manual content creation time by **85%**.
- Developed an interactive Streamlit web interface for real-time prompt customization and instant blog generation.
- Integrated LangChain Prompt Templates to enable dynamic, audience-specific content generation workflows.
- Applied advanced prompt engineering to enhance contextual relevance, coherence, and reduce API token usage.

Pneumonia Detection: CNN, VGG16, and ResNet | [Python](#), [Image Processing](#)

- Built a pneumonia detection model on chest X-ray images using TensorFlow, achieving **91% accuracy**.
- Compared CNN, VGG16, and ResNet architectures to optimize classification metrics and performance.
- Applied advanced image preprocessing with OpenCV (normalization, augmentation) to enhance model generalization.
- Fine-tuned hyperparameters and applied deep learning techniques to improve robustness across diverse datasets.

Achievements

Graduate Research Assistant: Selected from a competitive pool of over 150 applicants at University of Illinois.

HackSRM Hackathon: Won 1st Prize for presenting innovative solutions and technical excellence.

Top Hat Hackers: Secured 3rd Prize at UIS for developing an project supporting neurodivergent needs in workplaces.

Leadership: Led the Data Science Club at SRM AP, representing SRM innovations at various tech exhibitions and forums.